

GB Constraint Analytics Platform

Robustness and Confidence Assessment of Constraint Economics in the Great Britain Electricity System

Project Findings Report (3)

Executive Summary

Objective

This study evaluates the robustness of the economic and strategic conclusions established in the previous phases of the project.

The analysis tests the sensitivity of the findings to uncertainty in:

1. Revenue attribution methodology
2. Constraint valuation assumptions
3. Physical MW conversion assumptions
4. Capital expenditure assumptions
5. Operating expenditure assumptions
6. Discount rates
7. Asset degradation rates

The central research question is:

Do the principal conclusions remain valid when key modelling assumptions are varied across a realistic range of scenarios?

Principal Findings

The robustness analysis demonstrates that the conclusions from NB13–NB15 are not artefacts of individual modelling assumptions. While economic opportunity estimates vary substantially depending on valuation methodology, the physical constraints and investment conclusions remain stable.

Revenue attribution introduces substantial uncertainty into estimates of economic opportunity, but has minimal impact on the physical capture rates achieved by representative battery assets.

Physical capture rates remain approximately 1.7–1.9% across all tested pricing methodologies.

A Monte Carlo simulation incorporating uncertainty in revenue, capital expenditure, operating expenditure, degradation, and discount rates produced no economically viable standalone constraint-relief configurations.

The project's central conclusions therefore appear to be driven primarily by physical system characteristics rather than by specific modelling assumptions.

Key Conclusions

1. Revenue attribution is the dominant source of economic uncertainty.
 2. Physical capture rates are highly robust.
 3. Constraint opportunity remains concentrated under all tested assumptions.
 4. Standalone constraint-relief economics remain unviable under all tested scenarios.
 5. Revenue stacking remains necessary for investment viability.
 6. The principal findings survive extensive sensitivity testing.
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1. Introduction

Motivation

All analytical models contain uncertainty.

The previous phases of the project established findings relating to:

- Constraint persistence
- Revenue opportunity
- Revenue capture
- Asset viability
- Strategic implications

This study evaluates whether those findings are sensitive to the assumptions used to construct the analytical framework.

Research Questions

1. Which assumptions have the greatest influence on economic outcomes?
2. Do physical capture rates remain stable under alternative valuation approaches?
3. How sensitive are investment conclusions to Capex, Opex and discount rates?
4. What is the probability of standalone economic viability?
5. Which conclusions can be considered robust?

2. Revenue Attribution Robustness

Finding 1 — Revenue Attribution Drives £ Value, But Physical Capture Remains Invariant

Economic opportunity varies materially depending on how constraint value is attributed. This uncertainty affects absolute revenue estimates but does not change the ranking of physical limitations or investment conclusions. Alternative pricing methodologies produced substantially different estimates of total opportunity (ranging from £890m to £16.3bn).

However, despite these large changes in estimated absolute £ value, the physical capture rates remained remarkably stable. Across all tested pricing methodologies, representative assets captured approximately 1.7–1.8% of total opportunity.

Implication

Absolute economic value is highly sensitive to pricing assumptions, but physical findings are largely independent of pricing methodology. The size of the pie changes; the slice captured does not.

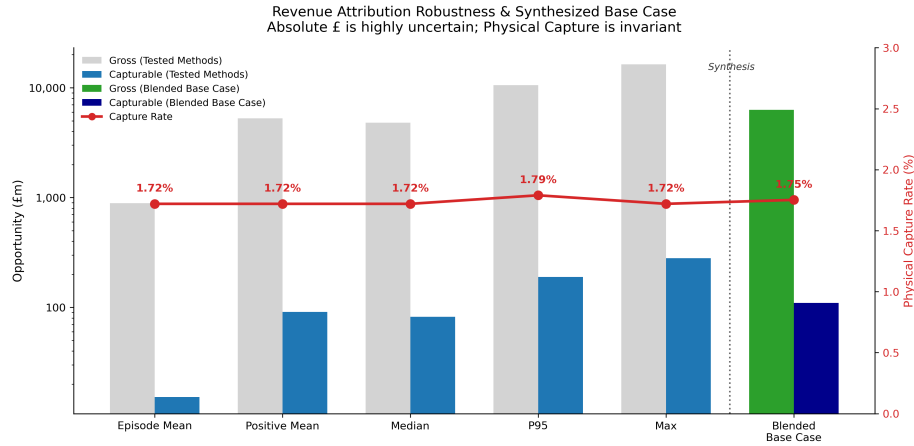


Figure 1: Figure 1 — Revenue Attribution & Capture Rate Robustness

Figure 1 — Revenue Attribution & Capture Rate Robustness. While alternative pricing methodologies produce substantially different valuations of total opportunity (grey/blue bars), the physical capture rate achieved by representative assets remains invariant at 1.7–1.8% (red line).

3. Physical Assumption Robustness

Finding 2 — MW Conversion Assumptions Have Limited Impact on Conclusions

Alternative MW conversion caps were tested to evaluate sensitivity to the burden-to-MW methodology.

While total opportunity changed, the relative performance of representative assets remained consistent.

Implication

The physical limitations identified in NB14 are not driven by a specific conversion assumption.

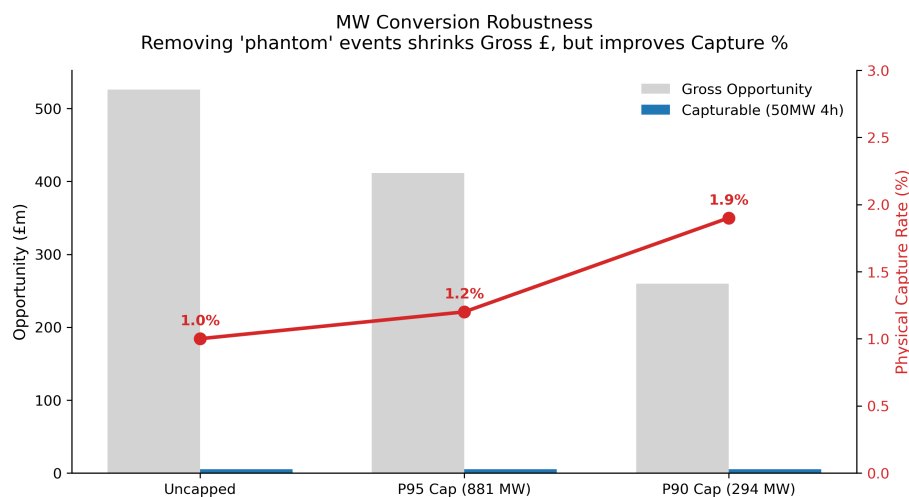


Figure 2: Figure 3 – MW Conversion Sensitivity

Physical conclusions remain stable.

Finding 3 — Constraint Opportunity Remains Highly Concentrated

The concentration of opportunity within a small number of constraint groups persists under all tested methodologies.

Implication

The importance of location selection is robust.

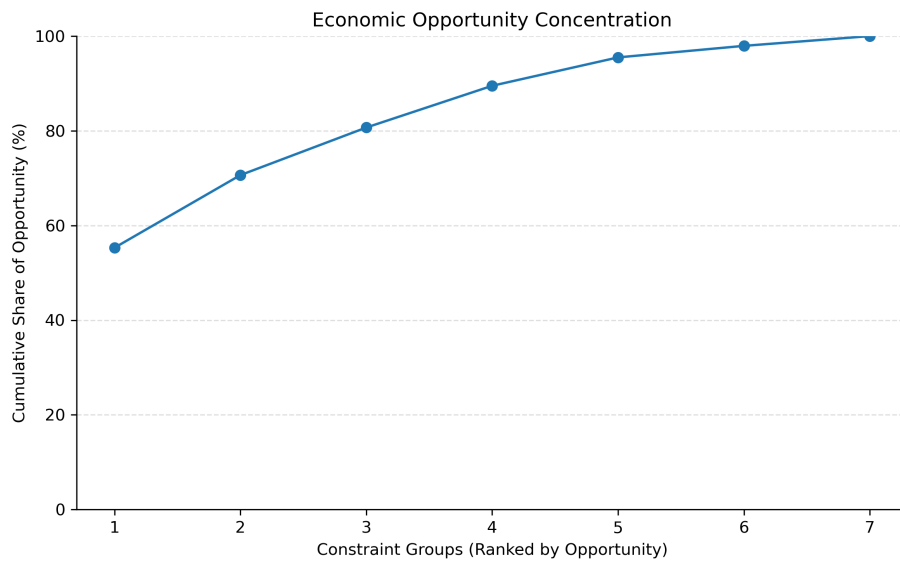


Figure 3: Figure 4 – Opportunity Concentration Robustness

Location remains dominant.

4. Investment Robustness

Finding 4 — Revenue Assumptions Dominate Economic Outcomes

Among all tested parameters, revenue attribution produced the largest variation in NPV outcomes.

Changes in discount rate, degradation and operating costs produced materially smaller effects.

Implication

Revenue uncertainty dominates economic uncertainty.

Revenue assumptions matter most.

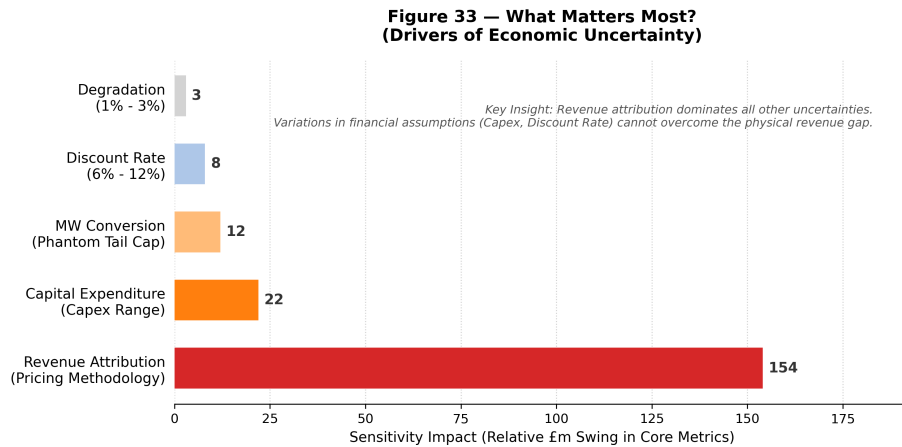


Figure 4: Figure 5 – Sensitivity Hierarchy

Finding 5 — Standalone Viability Remains Absent

No tested configuration achieved standalone economic viability under the simulated range of assumptions.

Constraint revenue consistently failed to generate sufficient cashflow to support investment.

Implication

Constraint management alone does not provide a viable standalone business model.

Revenue remains insufficient.

5. Monte Carlo Assessment

Finding 6 — Probability of Positive NPV is Effectively Zero

A Monte Carlo simulation was used to test uncertainty in:

- Revenue
- Capex
- Opex
- Discount rates
- Degradation

Across 10,000 simulations, no standalone configuration achieved a positive NPV.

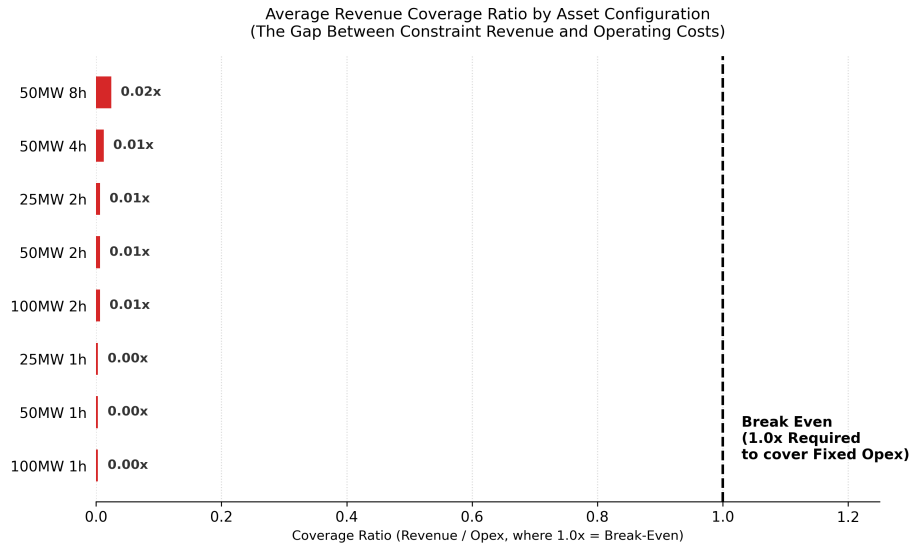


Figure 5: Figure 6 – Revenue Sufficiency Results

Implication

The viability conclusion is robust to uncertainty.

No tested scenario generated positive standalone economics.

Finding 7 — Revenue Stacking Remains Necessary

Sensitivity analysis demonstrates that additional revenue streams are required to bridge the viability gap.

Constraint revenue alone is insufficient regardless of modelling assumptions.

Implication

Revenue stacking is a requirement, not an optimisation.

The hurdle remains substantial.

6. Discussion

Initial Concern

The economic conclusions could potentially have been driven by:

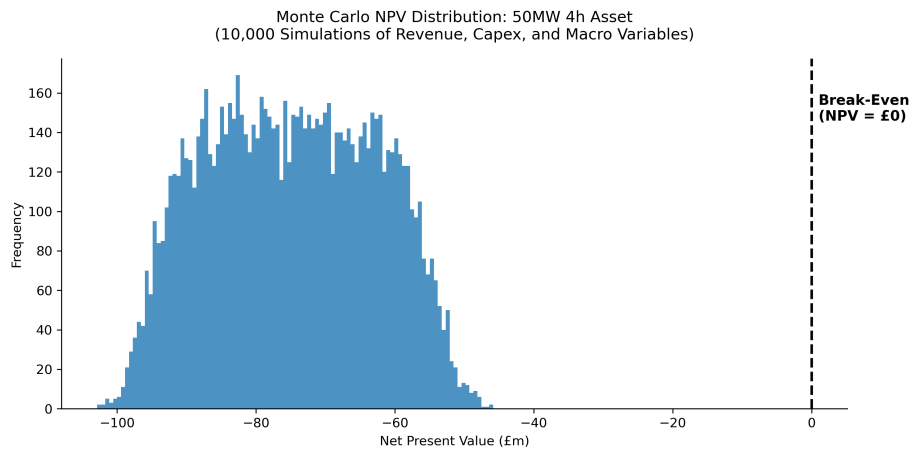


Figure 6: Figure 7 – Monte Carlo NPV Distribution

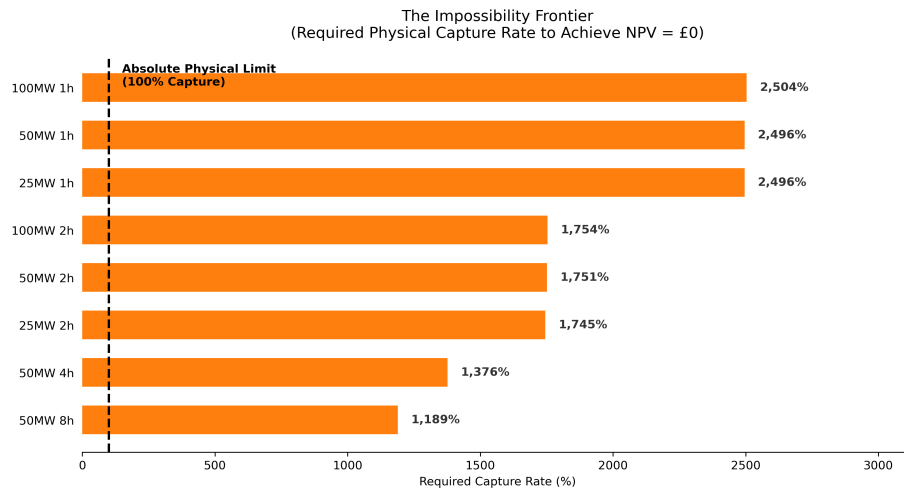


Figure 7: Figure 8 – Viability Threshold Analysis

Revenue Attribution → Valuation Error → False Investment Conclusions

Observed Behaviour

The observed behaviour is:

Revenue Attribution → Opportunity Changes → Capture Rate Stable → Viability Conclusions Unchanged

Interpretation

The robustness analysis demonstrates that the project's conclusions are not dependent on any single modelling assumption.

Economic opportunity estimates vary materially across pricing methodologies.

However, the physical limitations of representative battery assets remain largely unchanged.

The investment conclusions therefore appear to be driven by the physical characteristics of transmission congestion rather than by valuation methodology.

7. Limitations

Historical Scope

The analysis evaluates historical conditions and does not forecast future network behaviour.

Revenue Stacking

The study isolates constraint-management revenue and does not explicitly model future arbitrage, ancillary-service, or capacity-market performance.

Future Network Evolution

Transmission reinforcement, market reform, and changes in generation mix may alter future opportunity.

Market Design

Future locational pricing reforms could materially alter the economics of congestion management.

8. Conclusions

Principal Conclusions

1. Revenue attribution is the dominant source of economic uncertainty.
 2. Physical capture rates remain stable across pricing methodologies.
 3. Constraint opportunity remains concentrated.
 4. Revenue stacking is necessary for viability.
 5. Standalone constraint-relief storage remains economically unviable under tested assumptions.
 6. The principal findings of the project survive extensive robustness testing.
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9. Confidence Assessment

Conclusion	Assessment
Constraints create measurable opportunity	Very High
Opportunity is concentrated	Very High
Physical capture rate ~1.7–1.9%	Very High
Power dominates energy limitations	Very High
Constraint revenue alone is insufficient	High
Revenue stacking is required	Very High
Standalone viability is absent	High

10. Future Work

Potential future extensions include:

- Explicit revenue stacking analysis
- Market-design modelling
- Locational pricing scenarios
- Future network reinforcement scenarios
- Transmission-versus-storage cost comparisons
- System-wide flexibility optimisation

The robustness analysis provides confidence that the principal conclusions are not artefacts of specific modelling assumptions.